Krokhin Andrey, 2nd-year Bachelor's student at DRCT MIPT



Personal Information

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I'm a second-year bachelor's student at DRCT MIPT. During my first year, I completed the "System Programming and Compiler Technologies" course, which gave me hands-on experience with large-scale projects - from debugging and code analysis to comparing solutions based on execution speed and memory usage. Working closely with peers and mentors throughout the course also helped me develop stronger communication and teamwork skills.

Education

Moscow Institute of Physics and Technology, Department of Radio Engineering and Computer Technology. *Bachelor's, 2024 - 2029*
Specialization: Applied Mathematics and Physics.
GPA: 8.00/10.

Hard Skills

Programming languages: C, Python, x86 Assembler.
Tools: Git, IDA, Ghidra, GDB, Make, Perf.
Libraries/Frameworks: LaTeX, GraphViz, Matplotlib.
Languages: English (B1), Russian (Native).

Soft Skills

Communication, teamwork, adaptability, self-motivation.

Main Projects

Language (GitHub)

C | NASM

My programming language is specifically designed for mathematical analysis. It consists of three main parts: a lexer (tokenizer), a parser, and a compiler. The tokenizer converts the text into a list of tokens, and the parser converts this list into an n-ary tree, which the compiler then either translates into NASM assembly or compiles directly into an executable file.

Differentiator (GitHub)

C | GraphViz | LaTeX

A tool that uses a binary tree to differentiate a given mathematical expression and outputs the result to a LaTeX file. I used the GraphViz library to visualize the data and simplify debugging.

Hash Table (GitHub)

C | Perf | NASM | Python

A project focused on hash table optimizations. During the process, I analyzed the performance of various hash functions using the Perf profiler. It features three types of optimizations: assembly inserts, assembly functions, and C intrinsic functions.
